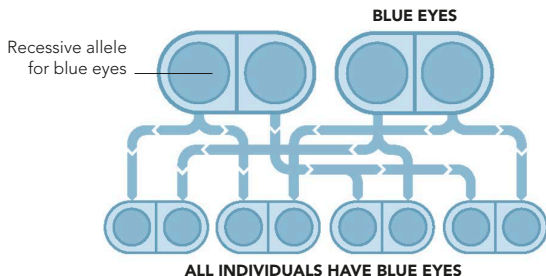


RECESSIVE AND DOMINANT GENES

Each gene in a cell exists in two versions, one inherited from each parent. In some cases these gene versions, or alleles, are different, and produce slightly different

results. One allele may be dominant and “overpower” the other, which is recessive. An example is eye colour, although this is not as simple as depicted below.

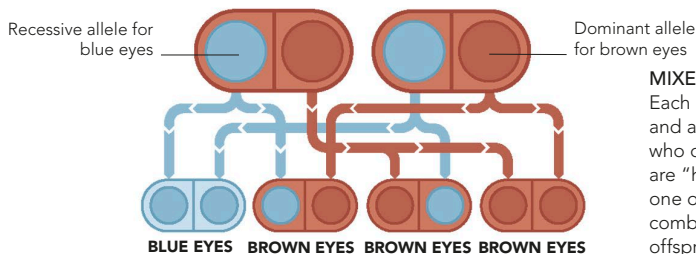
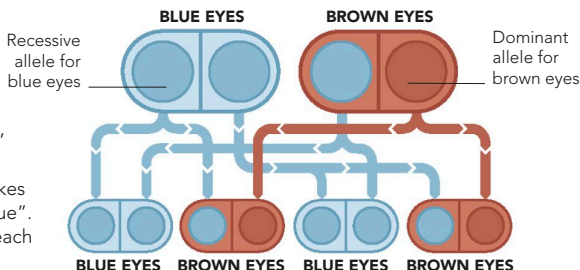


RECESSIVE AND RECESSIVE

Each parent has two alleles for eye colour. Here, both parents have only “blue” alleles. When both alleles are the same, the individual is said to be “homozygous”. Their children can only inherit “blue” alleles, and so all have blue eyes.

RECESSIVE AND MIXED

One parent has two “blue” alleles; the other, one “blue” and one “brown” allele. “Brown” is dominant and takes over when it occurs with “blue”. So the chance is 1 in 2 that each offspring has brown eyes.



MIXED AND MIXED

Each parent has a “brown” and a “blue” allele. Individuals who carry two different alleles are “heterozygous”. Only one of the four possible combinations leads to offspring with blue eyes.

DOMINANT AND RECESSIVE

One parent has two “blue” alleles; the other, two “brown” alleles. The four possible combinations all produce offspring with brown eyes, but all four still carry “blue” alleles.

